

RUENENBERG, Switzerland

WMO: 066450

Lat: 47.4344N

Lon: 7.8794E

Elev: 2001

StdP: 13.66

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.8	(c) 19.4	(d) 9.5	(e) 9.6	(f) 19.8	(g) 14.0	(h) 12.0	(i) 22.4	(j) 31.1	(k) 44.7	(l) 25.8	(m) 44.2	(n) 5.3	(o) 70	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.1	(c) 84.8	(d) 66.5	(e) 81.2	(f) 65.5	(g) 78.1	(h) 64.4	(i) 68.3	(j) 80.5	(k) 66.8	(l) 78.4	(m) 65.4	(n) 76.1	(o) 5.2	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 64.1	(b) 96.3	(c) 73.9	(d) 62.6	(e) 91.3	(f) 71.8	(g) 61.2	(h) 86.9	(i) 70.6	(j) 33.8	(k) 80.8	(l) 32.5	(m) 78.7	(n) 31.4	(o) 76.3	(p) 75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 22.0	(b) 16.4	(c) 13.1	(e) 12.9	(f) 89.6	(g) 5.3	(h) 3.0	(i) 9.1	(j) 91.8	(k) 6.1	(l) 93.5	(m) 3.1	(n) 95.2	(o) -0.7	(p) 97.4		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			12.0	71.5	5.5	1.5	8.0	72.6	4.8	73.4	1.7	74.2	-2.4	75.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	33.8	34.9	42.2	49.6	55.6	63.1	66.2	65.3	58.2	51.2	41.6	35.5
	DBStd	13.70	8.29	8.61	8.07	7.72	7.16	7.13	6.43	6.62	6.29	7.26	7.08	7.53
	HDD50	2118	504	423	261	102	27	2	0	0	8	76	264	451
	HDD65	5811	968	842	708	464	302	120	64	78	219	429	702	915
	CDD50	2061	1	1	19	89	200	394	503	475	253	113	12	1
	CDD65	279	0	0	0	1	10	63	102	88	14	1	0	0
	CDH74	1791	0	0	0	4	56	404	690	588	48	1	0	0
	CDH80	431	0	0	0	0	5	88	178	159	1	0	0	0
Wind	WSAvg	5.1	6.2	6.3	6.2	5.2	4.6	4.4	4.1	3.8	4.0	4.5	5.2	6.4
Precipitation	PrecAvg	43.0	2.9	2.7	2.8	3.2	4.2	4.4	4.6	4.3	3.5	3.4	3.5	3.7
	PrecMax	54.4	7.4	6.6	9.8	7.0	7.2	9.1	8.0	8.4	6.2	6.4	8.3	7.8
	PrecMin	29.6	0.6	0.9	0.9	0.4	1.3	1.0	1.5	0.7	1.7	0.4	0.2	0.0
	PrecStd	6.4	1.7	1.5	1.9	1.8	1.7	1.9	1.5	1.8	1.4	1.6	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	56.8	64.9	73.4	80.2	87.0	89.2	90.6	78.6	70.9	60.9	54.5
		MCWB	47.4	45.0	51.1	56.5	63.5	68.6	67.7	65.8	63.4	58.4	52.3	46.9
	2%	DB	51.3	52.5	60.6	69.2	74.5	82.5	84.8	84.3	75.0	66.9	57.3	51.1
		MCWB	45.3	44.1	47.9	53.8	60.3	66.3	65.9	66.7	62.9	57.4	50.3	45.7
	5%	DB	48.5	49.3	57.2	65.6	71.3	78.8	81.0	80.1	71.4	63.7	54.7	48.8
		MCWB	43.4	42.6	46.5	52.6	58.3	65.2	65.1	65.6	61.0	55.9	49.1	44.3
	10%	DB	44.9	46.5	53.9	62.0	67.7	75.1	77.5	76.5	68.1	60.9	51.7	46.3
		MCWB	40.9	40.7	45.2	50.6	56.7	63.4	64.0	64.3	59.7	54.8	47.3	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.8	47.1	52.1	57.6	64.7	70.9	70.0	70.0	66.1	60.6	53.6	48.6
		MCDB	53.3	52.7	62.1	70.3	76.6	83.4	83.7	82.2	74.9	67.7	58.4	52.2
	2%	WB	46.2	44.9	49.7	55.2	62.1	68.2	67.8	67.9	63.9	58.5	51.4	46.6
		MCDB	50.3	51.0	57.7	66.5	72.2	79.7	80.3	80.3	72.5	64.4	56.2	50.5
	5%	WB	43.6	43.1	47.8	53.3	59.8	66.1	66.3	66.5	62.1	57.0	49.4	44.8
		MCDB	47.7	49.0	55.0	63.5	68.5	76.7	78.3	78.2	69.4	62.5	53.9	48.5
	10%	WB	41.2	41.2	46.0	51.3	57.6	64.2	64.8	64.9	60.5	55.5	47.5	42.6
		MCDB	44.8	46.1	53.0	61.0	65.9	73.6	75.8	75.2	66.9	60.2	51.3	46.2
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.5	12.4	14.3	14.3	14.7	15.1	14.3	12.8	10.7	8.6	8.1
		MCDBR	11.4	13.6	17.0	19.3	18.8	19.2	20.0	19.5	17.2	14.3	12.7	11.4
		MCWBR	7.6	8.6	9.2	10.2	9.9	9.3	8.7	8.6	8.7	7.7	7.9	7.7
	5% WB	MCDBR	10.8	12.9	15.0	17.7	17.1	18.1	18.1	17.7	15.2	13.2	11.8	10.0
		MCWBR	7.7	8.6	8.6	9.8	9.4	9.2	8.5	8.5	8.2	7.8	7.9	7.5
Clear-Sky Solar Irradiance	taub	0.292	0.316	0.349	0.381	0.392	0.395	0.396	0.389	0.367	0.346	0.313	0.289	
	taud	2.465	2.398	2.353	2.289	2.298	2.326	2.315	2.350	2.394	2.443	2.484	2.469	
	Ebn at Noon	254	269	276	276	277	276	273	271	266	254	243	240	
	Edh at Noon	21	27	33	39	40	39	39	36	31	26	20	18	
All-Sky Solar Radiation	RadAvg	355	617	1005	1376	1594	1863	1789	1557	1176	738	399	289	
	RadStd	44	93	148	203	158	172	161	152	115	73	49	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	+0.56	N/A	N/A	N/A	N/A	N/A
Regional (117 neighbors)		N/A	N/A	N/A	+1.33	+0.74	+0.58	N/A	N/A	+99	+25

Nomenclature: See separate page